Safety-Efficiency Limits of Training-Free Whole-Prefix Visual Caching for Vision-Language-Action Inference

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Abstract

Vision-language-action (VLA) policies repeatedly encode similar observations during closed-loop manipulation, suggesting that visual features could be cached between policy queries. We study this idea through State-Aware Visual Refresh (SAVR), a training-free controller that reuses the complete projected visual prefix from a third-person and wrist camera while retaining current proprioception and the unchanged action head. The study uses a pinned OpenVLA-OFT checkpoint on LIBERO-Spatial and preserves 1,160 terminal episodes across a dense Full Refresh reference and three staged SAVR designs. Full Refresh succeeded on 100/100 calibration episodes. Nine initial SAVR settings achieved 0–52/100 success at 34.69–83.68% skipped visual refreshes; the least-degrading setting still lost 48 percentage points. Offline replay underestimated online reuse for every setting. For the least-degrading setting, 374 of 783 reuse queries concealed a per-camera threshold exceedance behind the two-camera mean, including 372 wrist-camera exceedances, and the first comparable reuse changed the action hash in all 87 analyzable episodes. Safety-constrained redesigns recovered success only by suppressing reuse: one configuration achieved 30/30 with zero reuse, while configurations at 6.72% and 10.57% skip achieved 29/30 and 27/30. A final post-hoc safety redesign achieved 69/70 at only 0.95% skip. No tested operating point satisfied its predeclared success-efficiency gate. These bounded negative results expose a closed-loop safety-efficiency frontier for whole-prefix caching and motivate sensor-granular or task-aware reuse rather than treating a multi-camera visual prefix as indivisible.

Keywords

vision-language-action models; robot manipulation; negative results; visual caching; adaptive computation; closed-loop control; inference efficiency

1 Introduction

Vision-language-action (VLA) models map language, images, and robot state directly to actions. RT-1 and RT-2 established scalable language-conditioned robot policies, while RT-X, Octo, and π_0 broadened cross-task and cross-embodiment policy learning [1, 2, 3, 4, 5]. OpenVLA released a 7B-parameter open model built from a Llama 2 language backbone and fused DINOv2/SigLIP visual features, and OpenVLA-OFT combined continuous actions, parallel decoding, and eight-action chunks to improve success and action-generation throughput [6, 7]. These advances make repeated visual processing an increasingly visible part of closed-loop inference.

Temporal redundancy offers an intuitive optimization opportunity. Adjacent robot observations often appear similar, and VLA-Cache has shown that selectively reusing minimally changed visual-token computation can accelerate VLA inference [13]. Subsequent methods have explored depth-guided compression, learned cache selection, model-confidence gating, and caching within iterative action generation [14, 15, 16, 17]. The difficult question is not whether redundancy exists, but whether a low-cost controller can identify when complete visual-prefix reuse is safe after its own actions begin changing the future trajectory.

We investigate the most direct version of this question: cache the complete projected prefix from both cameras and decide at each policy query whether to recompute it. State-Aware Visual Refresh (SAVR) combines image, robot-state, recent-action, and cache-age signals without changing policy weights. Our original hypothesis was that embodied signals would make whole-prefix reuse safer than visual similarity alone. The experiments did not support a useful operating point.

This paper reports the negative evidence rather than relaxing the gates or hiding failed variants. Across the tested OpenVLA-OFT/LIBERO-Spatial setting, permissive controllers changed actions and sharply reduced success, while conservative redesigns restored success by becoming nearly identical to dense inference. The study contributes:

1. a controlled, stage-gated evaluation of whole-prefix visual reuse spanning 1,160 preserved terminal episodes;
2. evidence that offline replay of dense trajectories underestimated the online reuse rate after cached actions changed the closed-loop trajectory;
3. a camera-specific diagnosis showing that averaging the scene and wrist views concealed 374 threshold exceedances in 783 reuse queries;
4. action-prefix evidence showing that the first comparable reuse changed the predicted action in 87/87 analyzable episodes; and
5. a documented safety-efficiency frontier: increasingly conservative controllers recovered success but reduced reuse to zero or less than one percent.

Our conclusion is deliberately bounded. We do not claim that visual caching is universally unsafe, that every reuse caused a failure, or that sensor-granular caching cannot work. We show that no tested *whole-prefix* controller met its predeclared gate on one pinned model, suite, and seed, and we identify concrete mechanisms that future caching systems should avoid.

2 Related Work

2.1 Vision-language-action policies

RT-1 demonstrated transformer-based control at scale, while RT-2 connected web-scale vision-language knowledge to robot actions [1, 2]. Open X-Embodiment pooled heterogeneous robot data and introduced RT-X models, and Octo used this data to train a flexible open generalist policy with multimodal, chunked action prediction [3, 4]. π_0 paired a pretrained vision-language backbone with a flow-matching action expert for general robot control [5]. OpenVLA made a 7B-parameter VLA and its training stack openly available [6]. OpenVLA-OFT adapted this model to continuous, chunked action prediction and reported strong LIBERO results [7]. We use the released OpenVLA-OFT stack and a pinned four-suite checkpoint as the fixed policy under study.

2.2 Efficient VLA inference and visual caching

VLA efficiency can be improved at several levels. TinyVLA and SmolVLA reduce model size and target deployment efficiency, while OpenVLA-OFT improves the action representation and decoder path [8, 9, 7]. EfficientVLA combines language-layer pruning, task-aware visual-token selection, and action-head feature caching [12]. These approaches complement general vision-transformer techniques such as dynamic token sparsification and token merging [10, 11].

Our study targets a narrower cost: recomputing visual features at every policy query. VLA-Cache is the closest published point of comparison; it compares adjacent visual inputs and reuses computation for selected minimally changed tokens [13]. DepthCache adds depth structure and motion-adaptive auxiliary-view compression, while a learned adaptive approach jointly selects cached tokens and predicts a reuse ratio [14, 15]. A recent preprint augments VLA-Cache with action-token confidence, and ActionCache instead reuses intermediate states in flow-based action generation [16, 17]. SAVR differs in granularity: it reuses the entire projected prefix of two cameras and employs hand-designed image, state, action, and horizon gates. This coarse boundary makes the negative result informative without contradicting successful sensor-, token-, or action-level reuse.

2.3 Closed-loop distribution shift

An offline replay assumes the observations and actions recorded under a reference policy remain representative after the controller changes. In closed-loop control, however, a changed action affects later observations, which affect later refresh signals and decisions. Our experiments directly measure the resulting gap between replay-calibrated and online reuse. This is not a general causal analysis, but it demonstrates why a cache controller should not treat dense-policy trajectories as a guaranteed online safety oracle.

3 Whole-Prefix Visual Reuse

3.1 Policy and cache boundary

At policy query t, the pinned OpenVLA-OFT model receives a fixed language instruction x, a third-person image o_t^{scene}, a wrist image o_t^{wrist}, and the current eight-dimensional robot state s_t. The visual towers and projector produce an ordered projected prefix

$$z_t = \left[z_t^{\text{scene}}; z_t^{\text{wrist}}\right]. \tag{1}$$

The unchanged downstream model and continuous action head predict an 8×7 action chunk,

$$\hat{A}_t = \pi_\theta(x, z_t, s_t). \tag{2}$$

Full Refresh (FR) recomputes both camera blocks at every query. Whole-prefix reuse instead supplies the last successfully refreshed prefix z_{τ_t} while still supplying current proprioception:

$$\tilde{z}_t = \begin{cases} z_t, & r_t = 1, \\ z_{\tau_t}, & r_t = 0, \end{cases} \qquad \hat{A}_t = \pi_\theta(x, \tilde{z}_t, s_t), \tag{3}$$

where $r_t = 1$ denotes refresh and $r_t = 0$ denotes reuse. Figure 1 shows the tested boundary. Reuse skips both visual towers and the projector; the language-model and action-head path remains unchanged.

3.2 SAVR 1.0 controller

Images were deterministically downsampled to 32×32 and normalized. SAVR 1.0 computed a mean absolute change for each camera relative to the last refreshed observation and then averaged the two

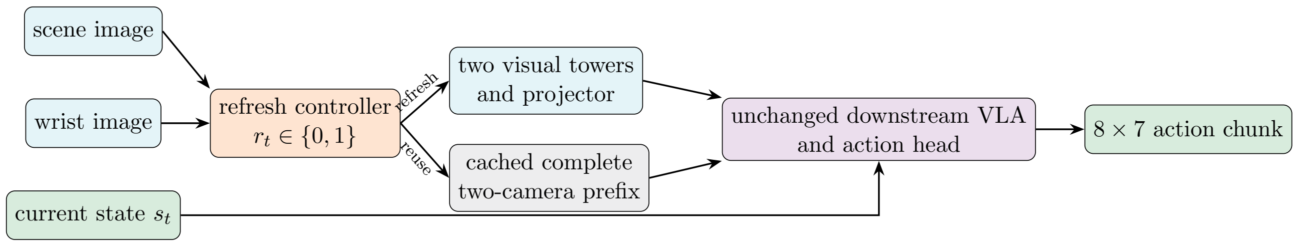

Figure 1: Tested whole-prefix cache boundary. A reuse skips both camera encoders and the visual projector but retains current proprioception and the complete downstream policy.

camera scores:

$$d_t^{\text{img}} = \frac{1}{2} \sum_{c \in \{\text{scene,wrist}\}} \frac{1}{N} \left\| D(o_t^c) - D(o_{\tau_t}^c) \right\|_1 . \tag{4}$$

Robot-state and action-chunk dimensions were normalized using calibration $q_{0.01}$ and $q_{0.99}$ bounds. The controller computed normalized root-mean-square changes d_t^{state} and d_t^{act}. It refreshed when the cache was empty, a signal was invalid, action history was incomplete, a score exceeded its threshold, or cache age reached horizon H:

$$r_t = \mathbb{I}\left[d_t^{\text{img}} > \gamma_{\text{img}} \;\vee\; d_t^{\text{state}} > \gamma_{\text{state}} \;\vee\; d_t^{\text{act}} > \gamma_{\text{act}} \;\vee\; h_t \geq H \right]. \tag{5}$$

For each target skip rate in $\{25, 50, 75\}\%$ and horizon in $\{2, 4, 8\}$, a fixed 0.001 quantile grid over Full Refresh traces selected thresholds whose replayed skip rate was closest to target, with conservative tie breaking. All thresholds were frozen before online execution.

3.3 Safety-constrained redesigns

Forensic analysis motivated SAVR 2.0. It replaced the two-camera average with independent per-camera local-change gates. Each 32×32 image was partitioned into an 8×8 grid, and the mean of the four largest patch changes formed each camera score. State was split into translation, orientation, and gripper groups; action was split into translation, rotation, and gripper groups. The controller added gripper-transition vetoes, five-query warm-up, two stable fresh queries before reuse, at most one consecutive reuse, and a hard episode-prefix reuse budget $b \in \{0.05, 0.10, 0.15\}$. A prospective reuse was rejected when it would make

$$\frac{R_t + 1}{t + 1} > b, \tag{6}$$

where R_t is the number of completed reuses before query t.

SAVR 3.0 was a transparently post-hoc localized redesign. Starting from the $b = 0.15$ SAVR 2.0 controller, it added a translation-direction-reversal veto and changed only the wrist-image threshold to 0.375. It was evaluated once on states not used in the SAVR 2.0 screen.

4 Experimental Design

4.1 Pinned system

Table 1 summarizes the fixed evaluation system. The base policy, checkpoint, prompt, preprocessing, action head, and simulator settings were not changed across refresh policies.

Table 1: Pinned evaluation system.

Base policy	OpenVLA-OFT, continuous L1 action head, eight-action chunks
Checkpoint	`openvla-7b-oft-libero-four-suite` Revision: `638918f3d1c2e43a39a8a20772bdb8b91835e4b7`
OpenVLA-OFT revision	`e4287e94541f459edc4feabc4e181f537cd569a8`
LIBERO revision	`8f1084e3132a39270c3a13ebe37270a43ece2a01`
Benchmark	LIBERO-Spatial tasks 0–9 [18]
Observations	Third-person image, wrist image, current 8-D proprioception
Hardware	One NVIDIA TITAN RTX (24 GiB) per run
Primary metric	Official terminal task success
Efficiency proxy	Fraction of policy queries that skipped both visual towers and projector

4.2 Populations and stage gates

The study followed immutable stage gates rather than repeatedly tuning on the same result.

- **SAVR 1.0 calibration:** 100 paired episodes per setting, covering tasks 0–9, initial states 0–9, and seed 0. Full Refresh used the same 100 task/state pairs. A setting was eligible only if all records reconciled, no technical failure occurred, no task collapsed to zero success when FR succeeded, and its paired success difference from FR was at least $-2\,pp$.
- **SAVR 2.0 Stage 1:** three predeclared candidates, each on tasks 0–9 and states 0–2 (30 episodes). Advancement required exactly 30/30 success, at least 2% online skip, zero technical failures, and complete cache/counter invariants.
- **SAVR 3.0 validation:** one frozen post-hoc controller on tasks 0–9 and states 3–9 (70 episodes). The gate required 70/70 overall, 7/7 per task, at least 5% skip, and zero technical or accounting failures.

The final holdout (states 10–49 and seeds 7, 17, 27) was never executed or inspected because no configuration passed its preceding gate. Matched-budget Periodic Refresh and Visual-Only Refresh baselines were also not run: the frozen protocol permitted them only after an eligible SAVR 1.0 configuration existed. Consequently, this paper reports a bounded method-rejection study, not final non-inferiority or superiority.

4.3 Correctness and integrity safeguards

Wrapped Full Refresh produced bitwise-identical actions to the unmodified OpenVLA-OFT path. Real reuse invoked zero vision-backbone and projector calls, executed the language-model and action-head path once, and used current proprioception. Every reported rollout terminated without infrastructure failure. Scientific failures were retained rather than rerun or relabeled. Run manifests recorded configuration and source revisions, and protected checkpoint files were restored to their original hashes. Large raw records remain indexed by authenticated summaries.

4.4 Measured optimization opportunity

Before testing reuse, a 50-episode Full Refresh pilot measured 662 steady-state queries after three predeclared warm-ups. Table 2 shows that the visual backbone and projector accounted for 15.874% of

Table 2: Steady-state Full Refresh component timing on 662 policy queries.

Component	Mean (ms)	Median (ms)	P95 (ms)
Policy-query CUDA	1267.44	1266.44	1287.80
Vision backbone	191.12	193.95	199.84
Visual projector	10.08	9.48	11.97
Backbone + projector	201.19	203.20	211.62
Residual downstream	1061.21	1058.79	1072.55

Table 3: Complete SAVR 1.0 online grid. Each row contains the same 100 paired task/state episodes as Full Refresh (100/100).

Configuration	Target skip	H	Success	Online skip	Difference
`s25-h2`	25%	2	52/100	34.69%	$-48\,pp$
`s25-h4`	25%	4	21/100	43.29%	$-79\,pp$
`s25-h8`	25%	8	23/100	47.28%	$-77\,pp$
`s50-h2`	50%	2	12/100	56.62%	$-88\,pp$
`s50-h4`	50%	4	4/100	63.26%	$-96\,pp$
`s50-h8`	50%	8	0/100	65.21%	$-100\,pp$
`s75-h2`	75%	2	3/100	64.14%	$-97\,pp$
`s75-h4`	75%	4	1/100	75.00%	$-99\,pp$
`s75-h8`	75%	8	0/100	83.68%	$-100\,pp$

query CUDA time. Even perfect visual elimination therefore had a measured ceiling of approximately 15.87% query-latency reduction before cache/controller overhead.

5 Results

5.1 SAVR 1.0 failed throughout the tested operating region

Full Refresh succeeded on all 100 paired calibration episodes and produced 1,309 signal traces. Table 3 reports the complete nine-setting grid. Every SAVR 1.0 setting missed the frozen $-2\,pp$ gate by a wide margin. The least-degrading setting, `savr-s25-h2`, achieved 52/100 success at 34.69% online skip, a paired difference of $-48\,pp$. Increasing target reuse or horizon did not reveal a safe region; two settings achieved zero successes.

Figure 2 places every staged operating point on a descriptive success-skip plane. Populations differ across stages, so the plot is not a randomized head-to-head comparison. Its purpose is to show the observed progression: high reuse accompanied low success, while high success required almost dense refresh.

5.2 Offline replay underestimated online reuse

All nine settings skipped more often online than predicted by replaying Full Refresh traces. For the 25% target family, replay predicted 24.98% skip, while online skip reached 34.69%, 43.29%, and 47.28% for horizons 2, 4, and 8, overshoots of 9.71, 18.31, and 22.30 percentage points. Once a reused prefix changed an action, later observations and controller inputs followed a different trajectory from the dense trace. The replay remained useful for screening thresholds, but it was not an online budget or safety oracle.

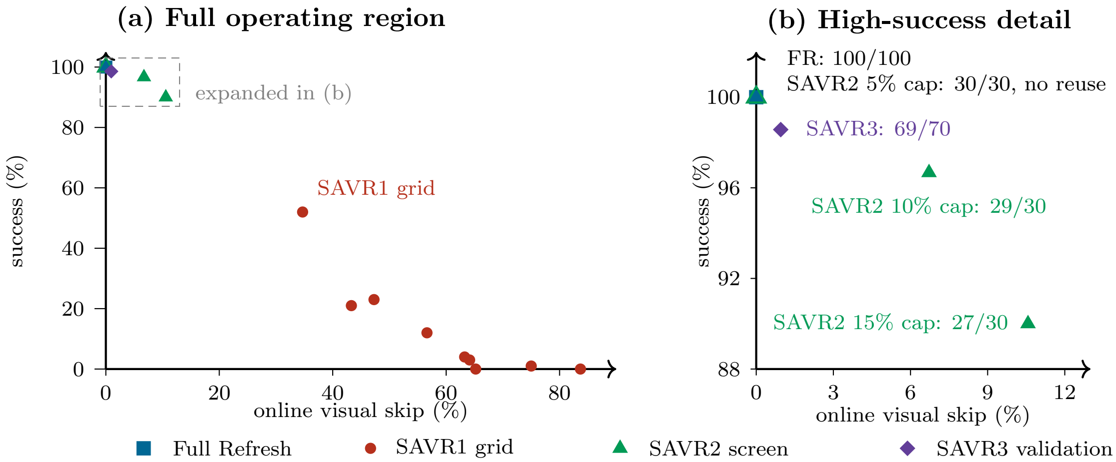

Figure 2: Observed success-efficiency frontier. Panel (b) expands the dashed region in panel (a) without moving any point. Full Refresh and the SAVR2 5% cap coincide at 0% skip and 100% success; the latter made no reuse decision. SAVR1 used 100 episodes per setting, SAVR2 used 30 per setting, and SAVR3 used 70. Because the stages used different populations and gates, the cross-stage frontier is descriptive rather than a randomized comparison.

5.3 Camera averaging concealed wrist dynamics

For `savr-s25-h2`, 374 of 783 reuse queries (47.77%) had at least one camera above the shared image threshold while the two-camera average remained below it. Of these concealed exceedances, 372 came from the wrist view and only two from the third-person view. This directly invalidates cross-camera averaging as a conservative gate in this setting. It does not prove that every wrist exceedance caused an episode failure.

5.4 The first reuse changed the predicted action

Eighty-seven `savr-s25-h2` episodes had comparable Full Refresh and SAVR action hashes through their first reuse. In all 87, the first mismatch occurred exactly at the first reuse query. Ten task-4 episodes already differed at query 0 and were excluded from this comparison; three episodes never reused. The result proves immediate policy-output divergence, but action hashes do not measure magnitude and cannot alone establish that the divergence caused terminal failure.

Reuse timing and streaks were also associated with outcome. Episodes whose first reuse occurred by query 2 succeeded 7/23 (30.4%); first reuse at queries 3–4 succeeded 18/42 (42.9%); first reuse at query 5 or later succeeded 24/32 (75.0%); and the three no-reuse episodes all succeeded. Every failed `savr-s25-h2` episode reached two consecutive reuse queries, permitting up to two eight-action chunks from stale visual features. These are exploratory associations rather than randomized causal effects.

5.5 Safety constraints collapsed toward Full Refresh

Table 4 reports the predeclared SAVR 2.0 screen and SAVR 3.0 validation. SAVR 2.0's most conservative candidate achieved 30/30 only by making no reuse decision. Candidates with useful skip rates missed the exact success gate. SAVR 3.0 recovered 69/70 success, but skipped only nine of 944 visual refreshes (0.9534%) and missed both its success and 5% minimum-skip gates.

The SAVR3 trigger audit explains the low reuse rate. Across 944 queries, wrist-image change triggered

Table 4: Safety-constrained redesign results. These were staged on different initial-state populations and are not pooled for inference.

Method	Episodes	Success	Queries	Reuses / skip	Frozen gate result
SAVR2-b05	30	30/30	388	0 / 0.00%	Failed minimum 2% skip
SAVR2-b10	30	29/30	461	31 / 6.72%	Failed exact success
SAVR2-b15	30	27/30	473	50 / 10.57%	Failed exact success
SAVR3	70	69/70	944	9 / 0.9534%	Failed exact success and 5% skip

279 times, while full-image change triggered only eight times. The two-stable-fresh requirement triggered 899 times, the prefix-budget rule 424 times, and translation reversal 412 times. Triggers can co-occur, so these counts are not additive. The controller did not fail technically; it conservatively refreshed on nearly every query.

6 Discussion

6.1 A closed-loop safety-efficiency frontier

The experiments reveal a consistent frontier in the tested region. Permissive whole-prefix reuse reduced visual calls but allowed stale representations to affect many future actions. Conservative rules prevented most risky reuse but also removed the intended efficiency benefit. This is not simply a threshold-selection failure: three redesign stages changed camera aggregation, local scoring, semantic grouping, temporal constraints, gripper vetoes, direction reversals, and hard budgets, yet no stage satisfied its own frozen gate.

6.2 Why offline calibration drifted

Dense-policy replay asks what the controller *would* have done on Full Refresh observations. Online reuse asks what the controller does after its previous cached action has already changed the robot state and observation stream. The 9.71–22.30 pp overshoot in the 25% family is direct evidence that these distributions differed. Future cache controllers should calibrate with bounded online data, model uncertainty under intervention, or enforce budgets that remain valid under trajectory shift.

6.3 Multi-camera prefixes should not be treated as indivisible

The scene camera was often stable while the wrist camera changed near the manipulated object. Averaging the two made a localized wrist change appear globally safe; independently gating both views then made whole-prefix reuse nearly vanish. This tension suggests a different abstraction: cache or refresh sensor blocks independently, or reuse tokens according to task relevance, rather than forcing both cameras to share one binary decision. Our study motivates this direction but does not experimentally validate it.

6.4 Skipped visual calls are not equivalent to end-to-end speedup

The component pilot limits the upside of any visual-only method on this stack. Visual backbone and projector work accounted for 15.874% of query CUDA time. SAVR3 skipped 0.9534% of those calls, corresponding to only about 0.15% of query CUDA time under a linear zero-overhead approximation.

Table 5: Primary evidence index. SHA-256 values authenticate the summarized artifacts.

Evidence	Repository location	Integrity identifier
SAVR1	`reports/PHASE6_CALIBRATION_REPORT.md`	FR trace input `c1724072...9804`
Diagnosis	`reports/PHASE6R_A_DIAGNOSIS_REPORT.md`	`9416014d...2ccf`
SAVR2	`reports/PHASE6R_D_STAGE1_REPORT.md`	summary `61a0c9dd...a9ebe`
SAVR3	`reports/PHASE6S_D_VALIDATION_REPORT.md`	summary `bb54eef3...d046`
Summary	`docs/evidence/negative_results_summary.csv`	tracked with source revision

We do not report this approximation as a measured speedup: matched end-to-end timing was not eligible after the success gate failed, and controller/cache overhead would reduce the benefit further.

7 Limitations

This study has several important limitations.

1. It evaluates one OpenVLA-OFT checkpoint on LIBERO-Spatial, one hardware class, and seed 0. Results may not transfer to other policies, camera geometries, tasks, or real robots.
2. The final holdout was intentionally untouched because no method passed the development gate. The results reject tested operating points but do not provide confirmatory population-level non-inferiority inference.
3. SAVR 1.0, SAVR 2.0, and SAVR 3.0 used different staged populations. Cross-version comparisons are descriptive, not randomized paired comparisons.
4. Matched-budget Periodic Refresh and Visual-Only Refresh were not run because the predeclared protocol required an eligible SAVR configuration first. We therefore cannot claim superiority or inferiority relative to those baselines.
5. SAVR3 was motivated by post-hoc failure analysis. Its one-time states 3–9 evaluation was protected from further tuning, but it is not independent confirmation of a prespecified original method.
6. Action hashes identify divergence but not action magnitude or causal responsibility for task failure. Raw SAVR 1.0 videos and contact-phase annotations were unavailable for a definitive failure taxonomy.
7. We tested complete-prefix caching. The findings do not rule out camera-block, token-level, learned, uncertainty-aware, or model-native caching.

8 Reproducibility and Evidence Integrity

The accompanying repository contains the controller, cache adapter, frozen configurations, schemas, analysis scripts, tests, decision ledger, and claim-indexed reports. Table 5 lists the principal evidence records. Large immutable run directories remain project-local and are indexed by their manifests and summary hashes.

All terminal task failures were retained. No threshold, success margin, or population was relaxed after observation. Technical failures were separated from scientific outcomes. Protected checkpoint files were restored, and no final-holdout outcome was inspected. These safeguards are central to the contribution: the negative result is useful only if unsuccessful trials and stopping decisions remain visible.

9 Conclusion

Training-free whole-prefix visual caching did not provide a safe and useful operating point for the tested OpenVLA-OFT/LIBERO-Spatial system. Aggressive reuse changed actions and sharply reduced success; progressively stronger safeguards recovered success only by collapsing toward Full Refresh. Offline dense-trajectory replay underestimated online reuse, and a two-camera average concealed frequent wrist-camera changes. These findings do not invalidate visual caching broadly. They show that closed-loop trajectory shift, sensor-specific dynamics, and cache granularity must be treated as first-class design constraints. Reporting this frontier prevents a failed coarse-grained design from being mistaken for evidence that visual redundancy is automatically safe to exploit.

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